



Generative AI, large language models, and their agentic framing in news media

Dennis Nguyen¹ · Magdalena Wischniewski²

Received: 20 February 2026 / Accepted: 17 April 2026
© The Author(s) 2026

Abstract

News reporting on generative artificial intelligence (GenAI) and large language models (LLMs) plays a central role in shaping the public epistemology of these technologies, as journalism co-produces public agendas and narratives. Central to these debates are framing practices that portray LLM-based technologies as mentalistic, often through metaphors that attribute human-like cognitive capabilities or even experiences to AI systems (e.g. AI “thinks” or “feels”). While anthropomorphic framing has a long history in technology discourse, such practices carry particular social, cultural, and political weight when they contribute to inflated perceptions of AI capabilities. The present study analyses LLM-centric news framing with a focus on mentalistic representations, examining their thematic contexts, valence, and metaphorical patterns from a comparative, cross-cultural perspective. It combines computational text analysis with manual content analysis to critically explore 18,032 English-language news articles published between 2022 and 2024 across 15 outlets in Europe, Asia, and North America. The findings show that mentalistic framings are present but not uniformly dominant, varying across editorial cultures and journalistic styles. Situated against news media’s growing organisational and economic entanglement with GenAI, the study contributes to debates on AI imaginaries, public epistemology, and journalism’s role in fostering critical AI literacy.

Keywords Large language models · Anthropomorphism · News framing · Artificial intelligence · News media analysis

1 Introduction

Current public debates on large language models (LLMs) that drive generative AI (GenAI), such as ChatGPT or Gemini, illustrate the continued newsworthiness of AI. These technologies make headlines on a daily basis and technology-centric journalism plays an important role in shaping public perceptions of AI (Cools et al. 2022; Zeng et al. 2022; Lewis et al. 2025; Kotliar 2026). However, critics of AI news discourses identify various risks in what they consider potentially harmful hype, especially regarding the use of anthropomorphic framings uncritically adopted from technology businesses (Bender and Hanna 2025): that is, the portrayal of LLM-based technologies as mentalistic

and virtually humanlike in their alleged cognitive capabilities (Schweitzer and Waytz, 2020). These framing practices carry particular social, cultural, and political weight if influential technologists suggest that LLM-powered AI exceeds human cognition. Research shows that language use around technologies affects how humans think about and interact with digital agents, including the trust placed in them (Langer et al. 2022). At the same time, reverting to metaphors to make sense of physical and social environments is just natural to how humans deploy language (Lakoff and Johnson 1981) in culturally dependent ways (Kövecses 2004). The use of metaphors is pertinent to journalism, which aims to convey condensed information about complex issues in formats that are understandable and appealing to broad audiences (Krennmayr 2014).

Arguably, not all forms of anthropomorphic AI framing serve the same argumentative goals, and cultural differences greatly matter in language use around, and perceptions of, such technologies (Fischer et al. 2021; Liebherr et al. 2025; Rahman et al. 2025). While a handful of studies investigate the use of anthropomorphic AI representations in the media, including a focus on LLM-powered GenAI (e.g. Bunz and

✉ Dennis Nguyen
d.nguyen1@uu.nl

Magdalena Wischniewski
magdalena.wischniewski@tu-dortmund.de

¹ Utrecht University, Utrecht, Netherlands

² TU Dortmund University, Dortmund, Germany

Braghieri 2021; Roe and Perkins 2023), more research is needed to scope the different modalities through which such framings are evoked in journalistic texts and on their potential consequences for public perception of emerging technologies. The present study builds on this by providing a critical exploration of the anthropomorphic framing of LLMs in global news media as mentalistic–agentic, i.e. when LLMs and associated GenAI systems are portrayed as having human mental capabilities and experiences. Methodologically, the empirical investigation utilises a mixed computational and manual content analysis of a large corpus of English news articles ($N=18,032$) retrieved from 15 news outlets across Europe, Asia, the Middle East, and North America, spanning the years 2022–2024. The multi-step analysis considers the thematic context, valence, and anthropomorphic metaphors applied in news texts.

The primary aim is exploratory and diagnostic: to systematically examine how anthropomorphic framing is applied in news coverage of LLMs, particularly regarding widespread critique of such language. Rather than starting from fixed assumptions about what constitutes “good” or “problematic” coverage, the study seeks to explore the range of existing practices and to better understand how mentalistic–agentic framings are deployed across contexts. While prior research has examined AI coverage more broadly, the present study’s contribution lies in (1) a more fine-grained and comparative analysis of how AI technologies—specifically LLMs—are linguistically constructed through anthropomorphic framing, shifting the focus to the text-level constructions through which technology is made intelligible in news; (2) providing a basis for empirically engaging with ongoing normative concerns about anthropomorphic language, particularly whether it aligns with techno-enthusiastic narratives or also appears in critical and ambivalent contexts; and (3) adding a comparative dimension by assessing whether these framings are globally uniform or culturally differentiated, thereby helping to identify how cultural contexts shape the ways in which societies “talk AI into being” (Bareis and Katzenbach 2022) through the lens of journalism.

2 Previous research and conceptual framework

2.1 News framing, AI imaginaries, and public epistemology

News media framing practices are a critical factor in how societies perceive and discuss emerging technologies, as journalism contributes to the formation of sociotechnical imaginaries. Simply put, sociotechnical imaginaries are interpretative frameworks shaping expectations towards technology as well as its integration within concrete

sociocultural configurations, such as, e.g. communities, organisations, and societies (Bareis and Katzenbach 2022; Sartori and Bocca 2023; Jasanoff 2015). Sociotechnical imaginaries emerge in complex discourse formations, especially via dominant technology narratives and framing practices therein, and are loaded with ideology (Gutierrez et al. 2025). Previous research has shown how technology discourses on past digital transformation trends—such as the rise of personal computers, the Internet, or big data—have shaped sociotechnical imaginaries centred on inevitability, disruption, revolution, and liberalisation, as well as rather conventional capitalist motifs of efficiency and wealth generation (e.g. Kvasny and Truex 2001; Kelly 2009; Hakken 1993). Current AI hypes echo these tech-enthusiastic narratives, often linked to anthropomorphic metaphors underscoring allegedly exceptional capabilities (Bunz and Braghieri 2021). Accordingly, they shape currently dominant AI imaginaries in global discourses. This has concrete implications for public perceptions and political decision-making about emerging technologies. For example, Liu et al. (2023) show that consuming AI-related news can indirectly influence support for AI policy and contribute to AI literacy.

The role of news media is ambiguous. On the one hand, they can uncritically reproduce and amplify narratives promoted by technologists and entrepreneurs, contributing to hype and inflated expectations. On the other hand, legacy journalism has become attentive to negative social, economic, and environmental consequences of AI (Nguyen 2023). In both ways, news media contribute to public epistemology around AI—that is, how technological developments enter public discussion, how they are best understood and interpreted, and, crucially, who the legitimate voices are that shape these discourses, thereby suggesting validity and quality of contributions, and ultimately defining what “truth” means within public discourse. Brandmayr 2021: 48) proposes that

[p]ublic epistemologies are recurring, complex, and relatively coherent but potentially unstable cultural schemes that define how one is to distinguish truth from falsity, what is the nature of science, what kind of people or institutions can be trusted to provide reliable knowledge, and which describe in a sequential or plot-like form why ignorance and error exist in the world and what is to be done about them.

AI narratives circulating in public discourse and the ideologically charged sociotechnical imaginaries embedded within them ultimately compete for recognition as the legitimate public epistemology around AI. Through agenda-setting, framing practices, choices in representation, and commentary, news media serve as a critical site where specific sociotechnical imaginaries gain wider public visibility, making them potential drivers in shaping the dominant

public epistemology around AI—particularly in the early stages of how complex technological developments unfold in society.

2.2 Research on news framing of AI

“AI” is a vague umbrella term for algorithmic, data-driven automation systems. Rather than perceiving it as a fixed technological entity, AI may be best approached as a discourse, subject to change over time (Lammar et al. 2025), with public attention recently being centred on LLMs and GenAI. The growing media attention on AI has stimulated research into different aspects of its journalistic coverage (e.g. Ittefaq et al. 2025; Lammar et al. 2025; Jones et al. 2022; Crépel et al. 2021). Most studies apply forms of content analysis, often with a focus on English text corpora (Chuan 2023). In a scoping literature review, Chuan (2023) identifies several broader themes in AI news that emerged over time: benefit and risk, ethics, responsibility and accountability, competition and collaboration, magic and religion, anthropomorphism, and AI as a nation’s future.

Previous studies focus on AI more broadly and show that the volume of relevant news articles has grown over the years (Valderrama et al. 2025; Nguyen and Hekman 2024; Sun et al. 2020). Generally, there are thematic similarities but also notable differences between news outlets and the world regions in which they are located. For example, Ittefaq et al. (2025) analysed 38,787 AI-related articles from 12 English language newspapers (2010–2023). They find that AI reporting mostly covers business, education, security, media, innovation, regulation, competition, healthcare, and politics. In their sample, noticeable regional differences between the Global North and Global South emerged: while news outlets located in, e.g. the UK and the USA tend to report less about innovation, their counterparts in Asia rarely address regulation issues (ibid.). This resonates with other studies that observe AI news agendas predominately focus on economics, applications, and politics (Valderrama et al. 2025; Sun et al. 2020), while non-Western news seem less concerned with AI risks (Nguyen and Hekman 2022). Taking a global-comparative perspective is thus important for identifying how regional and local filtration processes lead to divergences in AI news discourses (ibid.). However, Valderrama et al. (2025) show for the Latin American context that international -and especially U.S.-based- technology companies dominate AI news, suggesting that elite organisations and countries have wider global visibility.

AI news coverage further varies in its construction of scope. Lammar et al. (2025) show in their qualitative analysis of German AI news that journalists tend to contextualise automation in two temporal-spatial dimensions: either as *AI-in-general* with an outlook on the future and a global angle or *AI-in-particular*, focused on concrete locations

of technology effects (ibid). The authors see limitations to nuance in both, as hype remains dominant in AI news. Relatedly, in a more domain-specific study, Shorey and Rodriguez (2025) investigate US news coverage of AI and robotics in the recycling industry (2014–2024), especially during the COVID-19 pandemic. They observe that news reporting is mostly concerned with efficiency and accuracy, while limitations and downsides of AI remain less emphasised (ibid). Critically, executive decision-makers enjoyed higher visibility than workers, indicating a media representation gap in AI discourses.

Fewer studies analyse LLM and/or GenAI news more specifically. Roe and Perkins (2023) conducted an inductive thematic analysis of 671 news headlines on ChatGPT in the UK. They observed a “paradoxical discourse” (ibid) swinging between AI promises and societal risks that echoes controversy and polarisation among technologists, AI researchers, and academic critics. Roe and Perkins (2023) argue that such polarised media presentation warrants critical scrutinisation as framing practices perpetuating a lack of nuance may shape public perceptions of AI.

Taken together, previous research highlights that AI technologies have gained news value and visibility on news agendas but that current framing practices need critical assessment against their specific context; culturally, politically, and regarding organisational as well as individual factors shaping journalistic practices. Moreover, despite nuance and risk-sensitive AI reporting in certain media discourses, a techno-capitalist master frame dominates most AI news stories, pre-determining the horizon of AI imaginaries. Importantly, most previous studies chart out discourses with an analytical focus on AI news agenda setting, emphasis framing, and media representation. While some studies critically unpack anthropomorphic metaphors in the news (Bunz and Braghieri 2021), the news portrayal of AI as mentalistic-agent has received less attention, especially for the more recent trend of LLMs.

2.3 Anthropomorphic AI framing

Anthropomorphism generally refers to the psychological capacity of attributing humanlike characteristics and properties to non-human agents such as animals, deities, or objects (Epley et al. 2007). This includes mental states such as consciousness, beliefs, motivations, and intentions, but also encompasses behaviour towards (e.g. being polite to a robot), and feelings about (e.g. empathy) such non-human entities (Epley et al. 2007). Theories around anthropomorphism suggest that humans anthropomorphise non-human entities to understand and predict their behaviour (Quesque et al. 2024). As such, mental state attribution is based on the assumptions that the entity in focus can act (has agency) and has experiences (thoughts and feelings) (Gray et al. 2007).

This process is central to human social interactions, enabling people to navigate complex social environments and interact with others effectively. Consequently, attributing mental states to an agent influences how individuals perceive and interact with the world.

Anthropomorphism is a pertinent feature in computer science and AI research discourses (Stahl 1995). Anthropomorphising of digital technologies has been studied for computers (e.g. Nass et al. 1993), robots (e.g. Blut et al. 2021; Roesler et al. 2021), and AI-enabled technologies (Li and Suh 2022; Wischniewski et al. 2022). In the case of AI, comparing technological systems to humans and applying metaphorical conceptual mapping (Lakoff and Johnson 1981) arguably facilitates understanding and interacting with digital technologies. Hence, anthropomorphising seems virtually inevitable when talking about AI. The very term “artificial intelligence” itself is an anthropomorphic metaphor that was introduced in 1956 to describe the pursuit of machines that could simulate humanlike intelligence; it has sparked controversy and polarisation ever since (Cordeschi 2007; Emmert-Streib et al. 2020). Besides the matter of (un)intended anthropomorphism, critics argue that, as a “suitcase word” (Minsky 2007), ‘intelligence’ encompasses too many different and connected meanings such as thinking and consciousness, while it is also associated with certain emotional experiences.

Importantly, questions and concerns have been raised as to how to adequately talk about AI (Ciriello et al. 2025). Central to many of these studies are concerns that using anthropomorphic language, particularly mentalising language (i.e. expressing mental states, such as “think”, “understand”, “know”, or “believe”), may lead to misperceptions of technology. The primary concern is that mentalising language would lead users to anthropomorphise AI in ways that cause them to forget what they are dealing with. For example, Shanahan (2024) suggests that mentalistic vocabulary such as “knowing” or “understanding” may obfuscate the underlying mechanisms of LLMs, leading users to forget that they are, in essence, engaging with a statistical program that does not possess mind-like capabilities. The bottom line of this critique is that labelling AI systems as “intelligent” is misleading, as they lack “true” understanding and consciousness.

Concerns further centre on epistemological and ethical consequences of anthropomorphic language, affecting both public discourse as well as researchers. For example, Ouchchy et al. (2020) systematically examined media portrayals of ethical AI issues. Their findings suggest that, while the overall discourse appeared realistic, ethical consequences, such as access to new technologies, were underrepresented. Additionally, describing AI with humanlike characteristics provokes perceptions of AI as moral agents, attributing responsibility to mere tools (Waytz et al. 2010), and allowing

for “sociotechnical blindness” (Johnson and Verdicchio 2017, p. 587), limiting users’ ability to understand humans’ role in AI development and deployment.

Empirical studies suggest that concerns about language use around AI are legitimate, as it can influence laypeople’s perceptions and evaluations of AI (Langer et al. 2022). Likewise, Epstein et al. (2020) demonstrate that changes in language (i.e. “tool” or “agent”) can alter the way that people think about these systems by increasing perceptions of anthropomorphism and attributions of responsibility. Resonating with these results, Waytz et al. (2020) found that when autonomous vehicles were described in anthropomorphic language, participants trusted them more and allocated more responsibility to the car than to the driver. Moreover, developing the Mind Perception Dictionary, a tool to assess the attribution of mental states from how people talk, Schweitzer and Waytz (2021) found extensive evidence that the way people spoke about each other but also nonhumans (i.e. technologies) reflected differences in mental state attributions. These studies support the idea that using anthropomorphic language, particularly mentalistic language that implies agency, to describe GenAI and LLMs could affect media audiences’ cognition and behaviour towards these systems. Critically, related studies show that cultural differences further shape perceptions of AI in terms of application possibilities, capabilities, as well as associated benefits and risks (Rahman et al. 2025; Liebherr et al. 2025; Fischer et al. 2021).

Based on the above discussion, the present study aims to address the following research questions (RQs) with help of a computational content analysis centred on the news framing of LLM-driven GenAI technologies:

- RQ1: How do news media portray LLMs in their reporting about AI and to what extent do they evoke mentalistic -especially agentic- framings of the technology?
- RQ2: How do news framing practices differ between news outlets based on their cultural background?
- RQ3: To what extent does the news framing of specific LLM-based services and products differ?

For the present study, the mere mention of “AI” in the news does not, in itself, count as an anthropomorphic reference. This is a pragmatic choice. While “AI” is, strictly speaking, an anthropomorphic term, it is treated here as an established label for variety of technologies that are subjects of dynamic discourses. In the present case, analytical focus is placed on LLMs as one such subsumed branch of AI technology. The analysis (1) identifies the thematic contexts in the broader global AI discourse agenda; (2) identifies in which contexts and to what extent LLMs are the primary focus of AI news; and (3) examines how LLMs are described through anthropomorphic framings that use

mentalistic–agentic vocabulary. Considering previous research indicating the relevance of cultural differences in AI news reporting and AI perceptions, the study adopts a comparative lens for this exploration.

3 Data and method

3.1 Methods overview

The research design combines different natural language processing (NLP) methods for automated text analysis with manual content analysis. The NLP component includes topic modelling, word embeddings, dictionary-based text classification, and normalised word counts. The automated text clustering methods helped create a map of the broader AI news discourse, which then served to identify the specific thematic contexts or “emphasis frames” (Chong and Druckman 2007) in which LLMs are discussed.

Next, articles in which LLMs are presented as active subjects were collected in a subsample for manual content analysis, assessing whether such agentic framing is mentalistic (i.e. involving cognitive capabilities), experiential (i.e. involving emotions), or both (Schweitzer et al., 2022).

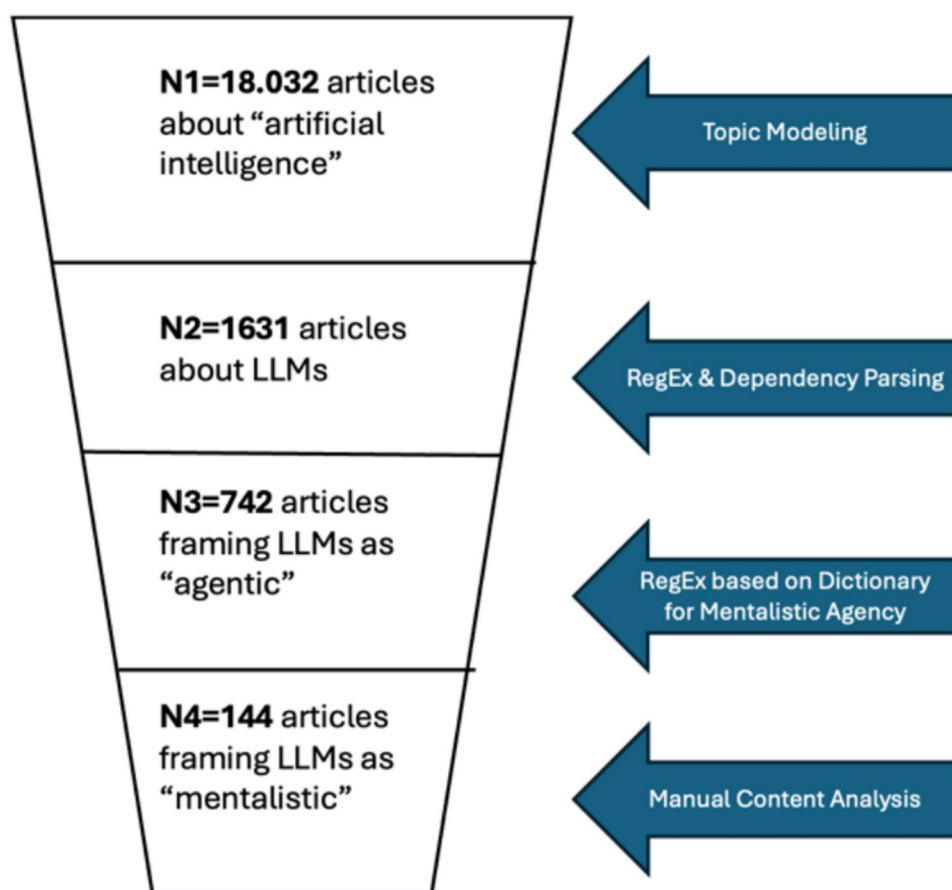
Representative examples highlighting observed framing trends were also retrieved at this stage.

In short, topic modelling provided a high-level structure of the corpus, while subsequent steps (rule-based filtering, dependency parsing, and manual coding) enabled increasingly fine-grained and controlled analysis. The entire process is visualised in Fig. 1.

3.2 Sample and data preprocessing

The dataset consists of $N1=18,032$ news articles about “artificial intelligence” and “AI” retrieved from 15 English-publishing news outlets, representing three world regions (Asia, North America, and Europe, covering the period between January 1st, 2022, and August 1st, 2024). The news articles were accessed via the Nexis archive. A critical limitation is that this search strategy does not include related terms such as “large language models” or “databases”. However, while it is possible for news articles to cover these technologies without mentioning “AI” or “artificial intelligence”, it is likely that most use these explicit AI keywords. Additionally, while the sampled news outlets differ in terms of cultural location, they represent distinct political-ideological

Fig. 1 Data processing



perspectives as well as journalistic genres within their respective media ecologies (Appendix 1).

Importantly, the dataset includes news articles covering the release of popular LLM-based GenAI services like ChatGPT and public reactions in subsequent months. The news articles were pre-processed in Python 3.9, using SpaCy's transformer-based English model (*en_core_web_trf*; Honnibal et al. 2020). The pipeline removed boilerplate text, stop words, and punctuation and lowercased all texts, before tokenisation and lemmatisation.

3.3 Automated content analysis

To chart the broader AI news agenda, topic modelling via BERTopic was applied to the full sample $N1=18,032$ (Grootendorst, 2022; model: *sentence-transformers/all-MiniLM-L6-v2*). BERTopic clusters text based on their semantic similarities and produces bags-of-words representations (BOWs) that are labelled by human annotators. For example, the BOW representation “cybersecurity, security, attack, scam, threat, fraud, phishing, vulnerability, malware, hacker” for text cluster 1 was labelled as the emphasis frame *Cybersecurity and Crime* (appendix 4). To test robustness in terms of the number of suggested text clusters and similarity of outputs, the BERTopic was run 20 times with different random seeds. The different trials ranged from 304 to 321 clusters, with a mean of 310.15. After manual inspection, the trial with the median of 316 clusters was selected for full human annotation and analysis. To further assess the reliability of the results, pairwise Jaccard similarity was calculated for the top word sets of matching clusters across the 20 trials. These values averaged 0.75, with most comparisons falling between 0.73 and 0.78. This indicates that, despite some boundary fluctuations typical of stochastic clustering, the model's overall thematic structure is sufficiently stable across trials.

Two researchers labelled the text clusters as emphasis frames applying an inductive approach inspired by systematic qualitative content analysis (Saldaña, 2021). The BOWs were treated as texts and primarily labelled based on their top words; when necessary, random examples from a given cluster were reviewed before assigning a label. After this initial annotation step, the two researchers compared and consolidated their labels. At this stage, “noise” clusters that could not be meaningfully labelled were discarded (12 of 316 clusters), resulting in an updated $N1=17,716$. “Noise” clusters typically consist of texts that, both in the BOW representation and upon manual inspection, do not share a clear thematic connection but are grouped algorithmically based on minimal statistical similarities. In total, only 1.75% of articles were removed in this way. This is standard practice in topic modelling workflows and highlights the inherent subjectivity involved in interpreting BOW outputs and

assigning labels to text clusters using such computational methods.

Next, to summarise the data more efficiently, the remaining 304 labelled clusters were manually grouped into 24 broader thematic meta-emphasis frames. Again, two researchers separately grouped the labelled clusters inductively, before comparing and collapsing their meta-emphasis frames labels. Afterwards, all texts in $N1$ were marked for their emphasis frame labels (cluster-specific and meta-grouping) as well as whether they include mentalistic–agentic framings of LLMs. This served then as the basis for quantitative exploratory analysis. Additionally, word embedding for the pre-processed texts were generated with Word2Vec (Mikolov et al. 2013) to zoom in on the semantic neighbourhoods around selected keywords to further probe the framing of LLMs.

3.4 Identification of LLM-centric articles

Articles that discuss LLMs specifically were filtered out from $N1$ with the help of an exhaustive keywords list (appendix 2) and a RegEx-based classifier. Human inspection of a random 10% of the classified articles indicated an accuracy of over 95% after inspection by a single researcher. This yielded $N2=1631$ LLM-centric articles, which is 9% of $N1$. Next, each LLM-centric article was segmented into sentences via a RegEx-based splitter to enable sentence-level analysis. This was necessary for isolating sentences in which LLMs are positioned as the subject. To retain immediate context without diluting precision, adjacent matching sentences were aggregated per article (preserving order) into contiguous spans. Using SpaCy's transformer model (*en_core_web_trf*), the sentences underwent dependency parsing, and a detector was implemented to mark sentences where LLMs appear as a nominal subject (nsubj/nsubjpass). This process yielded a subsample of 11,663 sentences, of which 2188 sentences had LLMs as a subject.

3.5 Manual content analysis

Two human coders categorised these sentences as agentic (y/n) to further filter out instances in which LLMs are presented as active subjects, with an acceptable inter-coder agreement score of $KALPHA=0.76$ for a random 10% of the LLM-as-subject-sentences. Sentences that frame LLMs as agentic were found for $N3=742$ articles in this way. Of these, 377 articles were filtered out as candidates for presenting LLMs as mentalistic- and/or experiential-agentic based on the dictionary compiled by Schweitzer et al. (2020). These were manually categorised for mentalistic and/or experiential following a simple coding scheme with an acceptable $KALPHA=0.78$ (appendix 3). This step yielded $N4=144$ articles where LLMs are portrayed as mentalistic,

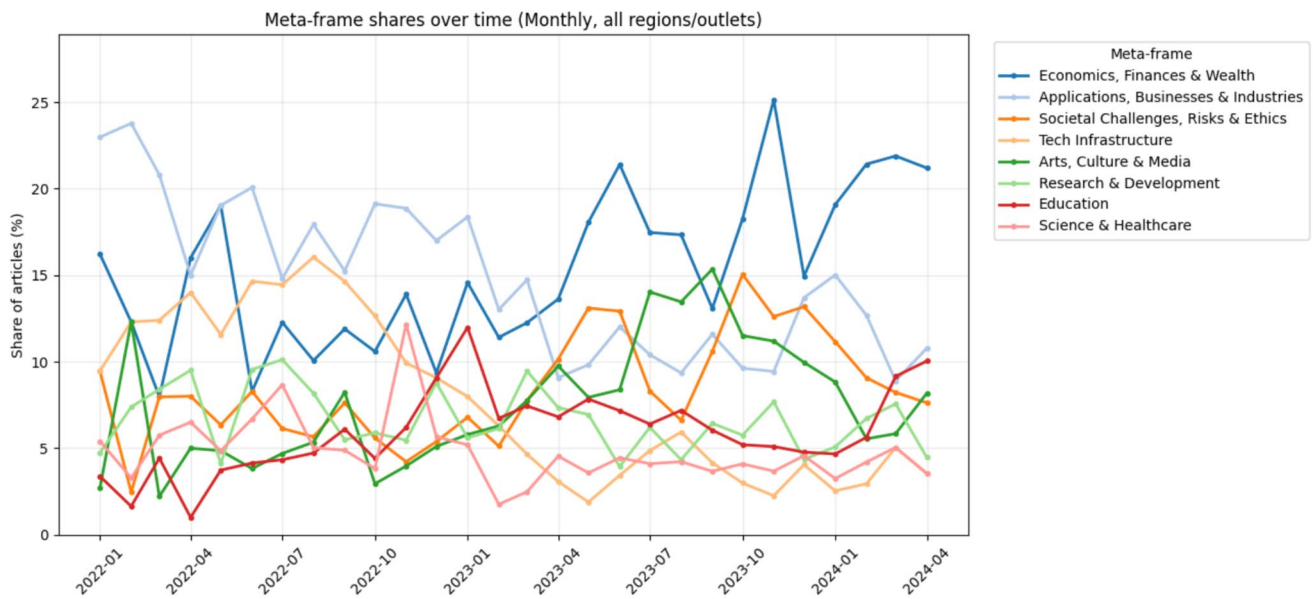


Fig. 2 Top eight meta-frames based on distribution in N1=17,716 over time

i.e. 8.8% of N2. No instances were identified where LLMs were portrayed as experiential. Certain metaphorical uses of mentalistic–agentic framings, such as “machine learning” may be considered part of the term’s semantic meaning, as is typical for many metaphorical expressions. This may be particularly the case for verbs such as “learning,” which are also used to describe the technological process itself (e.g. machine learning, deep learning). For this reason, the study adopted here a manual content analysis to minimise potential false positives, guided by a clear coding scheme and illustrative examples.

Importantly, the present analysis does not treat “AI” itself as an inherently anthropomorphic keyword; instead, it identifies linguistic constructions that attribute mental states or cognitive capacities to subsumed technological systems (e.g. “the LLM knows”). Such expressions are classified as anthropomorphic because they ascribe capacities typically associated with human agents, whereas “AI” can function as a comparatively neutral label. The approach thus prioritises explicit attributions of mental states as the core unit of analysis. Furthermore, the analytical approach captures anthropomorphic language irrespective of polarity, including cases where agency is explicitly denied (e.g. “the LLM does not understand”). To assess the impact of such negations, an additional analysis identified negated constructions using dependency parsing and aggregated results at the article level. Of the full corpus (N1=18,032), 1599 articles contain at least one instance of an LLM as a grammatical subject, of which only 99 articles (6.19%) include negated constructions. This indicates that explicit denials of agency constitute

a small minority of cases and do not drive the overall patterns observed.

4 Findings

4.1 Thematic emphases in general AI news coverage

While AI-related news articles frequently seem to dominate news agendas, they only form a fraction of all news output. For example, between January 2022 and April 2024, merely 0.9% of all *New York Times* articles (1980 out of 216,544) covered AI beyond mentioning it in passing.¹ However, the number of AI-related articles noticeably increased over time across outlets (appendix 4).

AI news reporting covers a wide spectrum of specific topics across societal domains and industries, though economic-commercial issues remain dominant (appendices 5 and 6). Notably, there are temporal shifts in the global AI news agenda setting (Fig. 2; appendix 7). For example, articles about concrete AI *applications, businesses, and industries* went down around spring 2023, while articles about broader *economics, finances, and wealth* developments went up. Both meta-frames illustrate the dominance of a technology-capitalist master frame, but this may signal a shift in which AI is perceived as a more general economic factor.

¹ This comparison draws on the total number of articles on any topic available in the Nexis Archive.

Furthermore, there is noticeable coverage of *societal challenges, risks and ethics*, which peaks in 2023. This meta-frame includes diverse negative AI news about, e.g. harmful effects on vulnerable groups (such as children), misinformation, climate effects, mass unemployment, or existential risks. Coverage of AI in the context of *education* spiked in November 2022, the month in which ChatGPT was publicly released. A similar increase of article share can be identified for *arts, culture, and media*, which includes coverage of GenAI implications for creative industries. Overall, the AI news discourse mostly focuses on economic, domain-specific, infrastructural, and industrial thematic contexts, while research- and security themes are also part of the global agenda. Despite some variation in the exact topic selection for emphasis framing between the different news outlets, the general AI news agenda looks similar across outlets (appendix 8).

Differences between outlets and their respective world regions mostly concern news reporting about AI risks and ethics (Fig. 3). These issues are more likely to be covered in the sampled European and, to a lesser extent, North American outlets, while their Asian counterparts predominately focus on the economic dimension and implications for education. Thus, all outlets share a similar primary emphasis on economics but diverge in their secondary emphases in their respective news agendas.

Investigating the relationship between an outlet's world region and their thematic focus in the respective meta-frame

distribution further supports this observation ($\chi^2(46, N = 17716) = 1399.64, p < 0.001$), with standardised residuals revealing distinct regional emphases: Asian outlets are more focused on utilitarian as well as developmental frames, such as *education* and *applications, businesses and industries*, while they are less likely to cover *arts, culture, and media* or *societal challenges, risks, and ethics*. For European outlets, the situation is almost the reverse, with a stronger emphasis on, e.g. *politics, governance*, and *societal challenges, risks and ethics* but comparatively less focus on *chip industries, education*, or *tech infrastructure*. North American outlets are characterised by a stronger emphasis on *arts, media and culture, tech infrastructure*, and *tech design* and less coverage of, e.g. *governance* and *education*. The results suggest that despite a globally shared techno-capitalist master frame, AI news coverage is likely shaped by local discourse regimes that are in turn configured by the sociopolitical structures shaping public- and media discourses.

4.2 News coverage of LLMs in AI news

LLM-centric news articles mirror some of these broader framing trends around AI more generally, but also show nuanced differences between news outlets. The overall volume of articles that focus on LLMs is relatively modest: N2=1631 were identified as such, which constitutes 9.2% of N1. Notably, there is considerable convergence across outlets in the temporal attention peak on LLMs across world

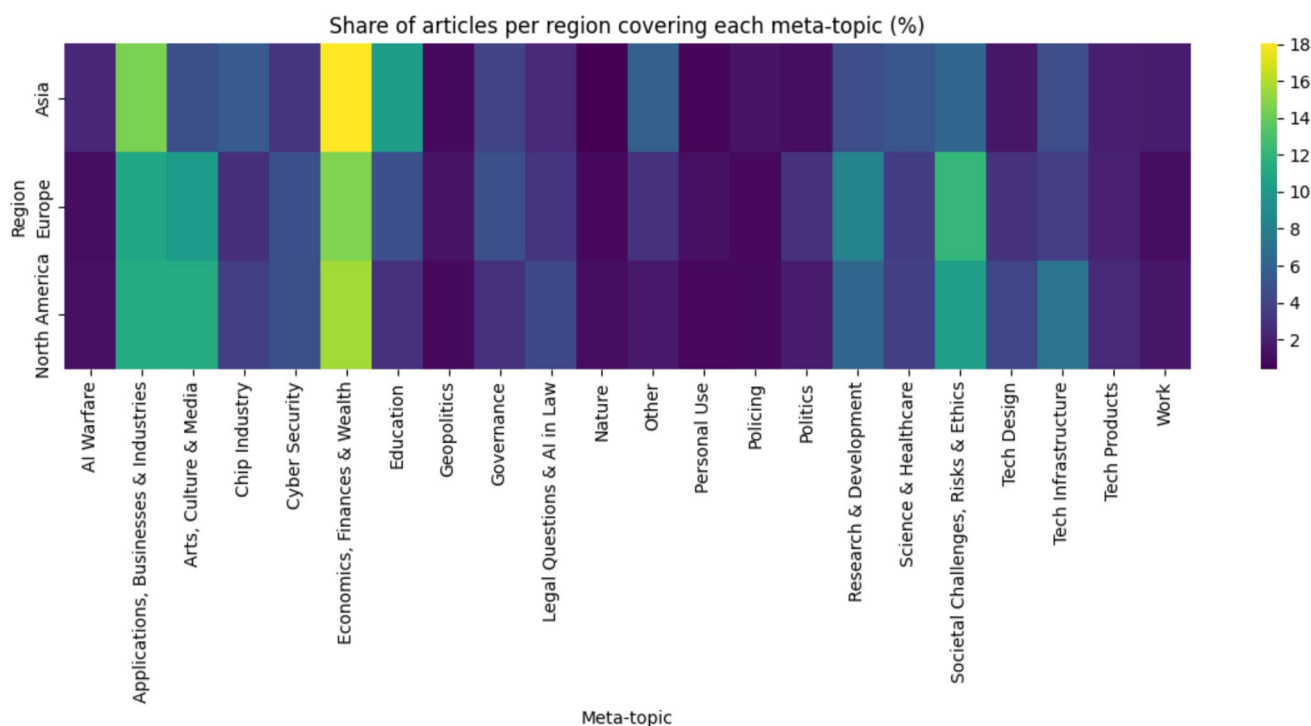


Fig. 3 Meta-frame distribution per world region in N1=17,716

regions (Fig. 4). All news outlets covered LLMs most frequently between late 2022 and early 2023. This suggests that especially the launch of ChatGPT captured global media attention and placed LLMs at the centre of the global AI imaginary, at least temporarily. However, the number of articles specifically covering LLMs decreased relatively sharply as of mid-2023, with fewer and less extreme peaks towards early 2024.

News articles about LLMs mostly cover specific GenAI services offered by companies such as OpenAI or Google. In fact, OpenAI's ChatGPT dominates LLM articles by being mentioned in 82.5% of $N_2=1631$, while its competitors are represented less frequently (15.7% for Google's Gemini/Bard, followed by 2.5% for Claude, and 1% for Mistral). Arguably, the widespread publicity and rapid user uptake of ChatGPT contributed to its news value.

There are some notable differences regarding the dominant meta-frames evoked in LLM-centric articles compared to AI more generally (Table 1). Over 13% of LLM articles focus on *Chatbots and tech design*, in which conversational capabilities and use cases are covered, while technological limitations are highlighted as well. This is followed by *research and development* (11.8%), which includes articles about language and computers, image generation, engineering problems, innovations as well as “hallucinations” and questions around LLM's and “consciousness”. Other notable meta-frames pertain to economic value and the commercialisation of LLM-driven GenAI (11.8% for *economics, finances, and wealth* and 11% for *applications, businesses, and industries*). Notably, LLM-centric articles cover the perceived implications for *education* (8.9%) as well as *societal*

Table 1 Top ten meta-frames in LLM-centric articles ($N_2=1631$)

Meta-frame	% of N_2
Chatbots and tech design	13.4
Research and development	11.9
Economics, finances, and wealth	11.8
Applications, businesses, and industries	11.0
Education	9.0
Societal challenges, risks, and ethics	8.7
Legal questions and AI in law	6.3
Arts, culture, and media	5.8
Cyber security	5.5
Tech infrastructure	3.5

challenges, risks, and ethics (8.6%) and *legal questions and AI in law* (6.3%). Again, regional differences emerge here in line with the earlier observation that Asian outlets are less likely to cover societal challenges and legal contexts than European or North American ones ($\chi^2(40, N = 1631) = 131.93, p < 0.001$).

45.5% of LLM articles present the technology as agentic, i.e. capable of actively “doing things” in general, and differences between world regions appear marginal with noticeable convergence in relative volume and temporal trajectories (appendix 9). However, a much smaller number of articles evoke mentalistic–agentic framing that quasi-equates these technologies with human cognitive capabilities (144 articles, i.e. 8.8% of $N_2=1631$ or 0.8% of $N_1=17,716$). Importantly, these are exclusively mentalistic–agentic in the sense of assigning human cognitive capabilities to LLM-driven

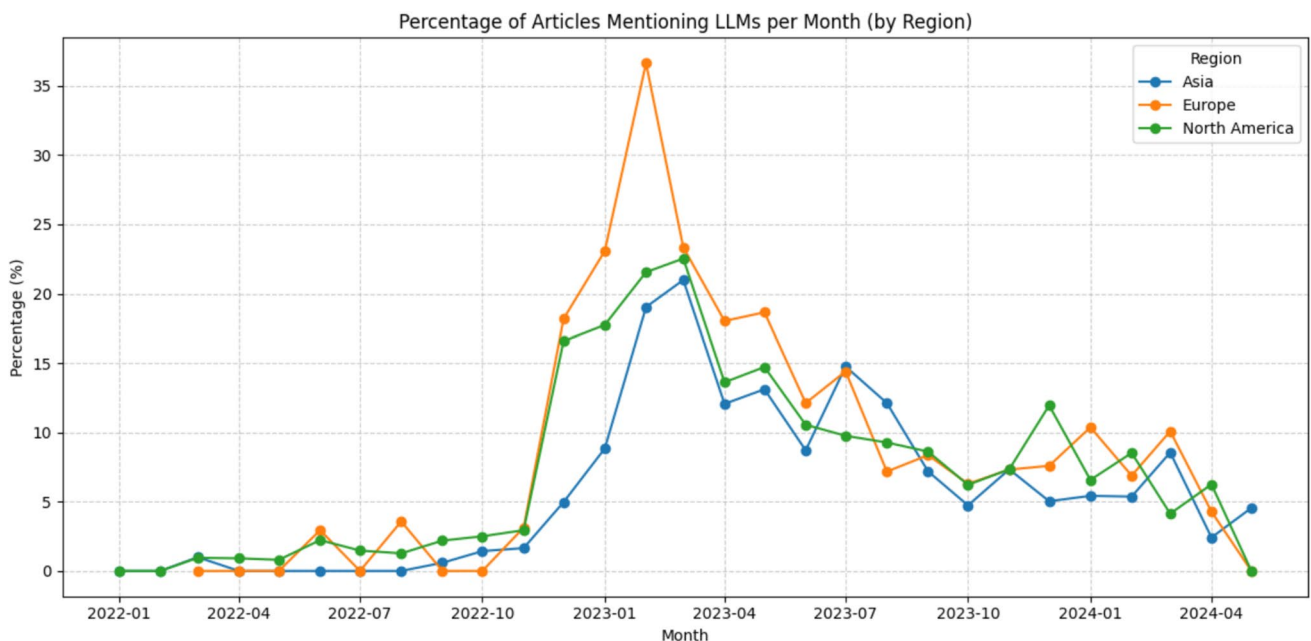


Fig. 4 Volume of LLM-centric news articles per world region over time for $N_2=1631$

GenAI as in being able to “think”, “reason”, “write”, or “talk”. No instances were identified in which these technologies were presented as having human-like experiences, such as undergoing emotional states (e.g. “feels”, “is happy”, “is sad”). There is a considerable overlap between outlets and world regions regarding when and to what volume articles tend to frame LLMs with mentalistic–agentic characteristics (Fig. 5).

The distribution of articles presenting LLMs as mentalistic–agentic differed significantly across specific outlets ($\chi^2[12, N = 17,716] = 74.00, p < 0.001$), although the effect size was small (Cramér’s $V = 0.06$). The *New York Times* ($n = 27$), *Straits Times* ($n = 24$), and *Agence France Presse* ($n = 13$) contributed disproportionately to the observed counts, while most other outlets had fewer than 10 articles. In contrast, no significant differences emerged across regions, $\chi^2(2, N = 17,716) = 0.58, p = 0.75$, Cramér’s $V = 0.01$. The distribution of such articles is relatively balanced among Asia ($n = 51$), North America ($n = 58$), and Europe ($n = 35$). This suggests that mentalistic–agentic framing is likely to be shaped by organisational-editorial and/or individual journalistic practices rather than broader cultural differences.

Mentalistic–agentic framings that portray LLMs from an anthropomorphic angle takes a variety of forms, aiming for different conceptual metaphorical mappings and sentiments. For example, some articles deploy an arguably more humorous framing with sometimes sarcastic undertones. The NYT article *A ChatGPT With Eyes and Voice* (Appendix 10A) is an example:

chatgpt-viral artificial intelligence sensation, slayer of boring office work, sworn enemy of high school teachers and hollywood screenwriters alike—is getting some new powers. on monday, chatgpt’s maker, openai, announced that it was giving the popular chatbot the ability to “see, hear and speak” with two new features. the first is an update that allows chatgpt to analyze and respond to images

On the one hand, LLM-driven GenAI is here metaphorically described as a quasi-anthropomorphic entity (“slayer” and “sworn enemy”) to allude to the challenges that these technologies pose for the specific sectors of education and creative industries. On the other, the article directly cites OpenAI’s own mentalistic–agentic framing for its products’ expanded functionalities to process audio-visual data. While the article outlines the technicalities of data processing to some extent, it continues to reproduce technologists’ anthropomorphic framing uncommented: “*The ultimate goal, according to some researchers, is to create an A.I. capable of processing information in all the ways a human can*” (ibid., our emphasis).

Arguably, the use of conceptual metaphors explaining technological features with human capabilities makes the aspired goals explicit and easy to understand (Lakoff and Johnson 1981). Yet, it simultaneously contributes to a structural anthropomorphisation of AI. Other articles explicitly anthropomorphise GenAI by comparing it to humans in an unambiguous way, as shown in the headlines *ChatGPT is only 2 months old, but has a brain of a 9-year-old*

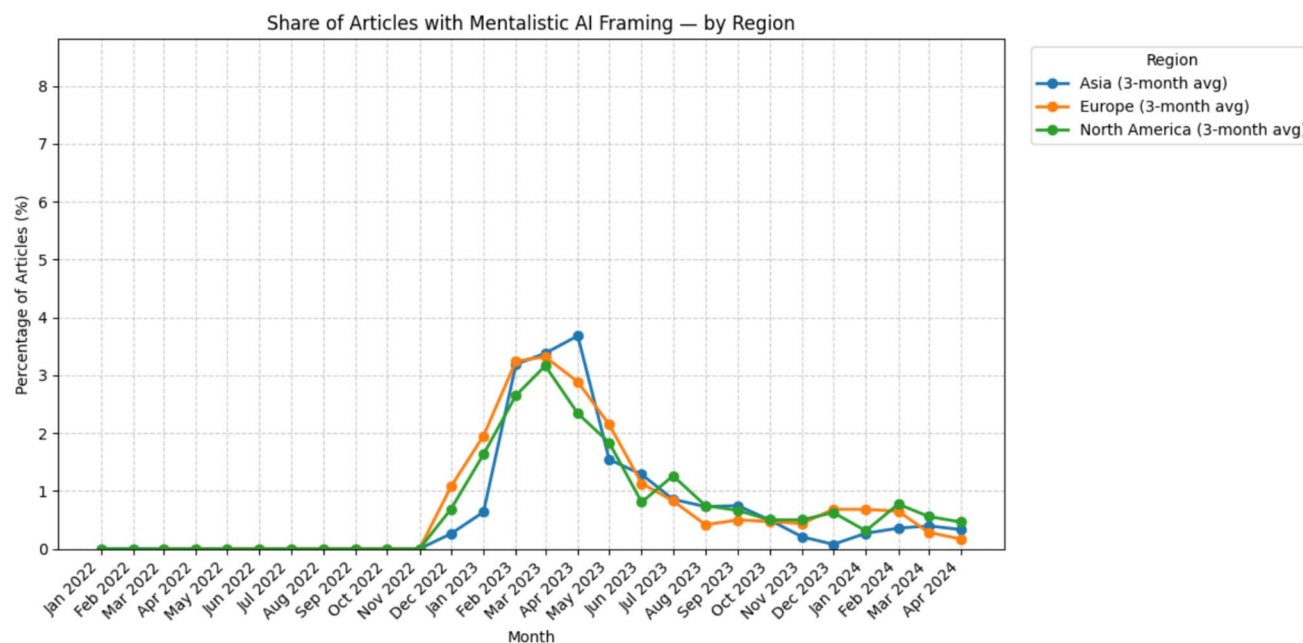


Fig. 5 Distribution of articles framing LLM-driven GenAI as mentalistic–agentic over time, per world region (N1=17,716)

(Appendix 10B) or *AI chatbot gives right answers on journalists' jobs* (Appendix 10C). Other forms of anthropomorphising LLM-driven GenAI take a more negative tone. For example, the TechRadar article *ChatGPT is leaking... again—we shouldn't be surprised but we should be* (Appendix 10D) frames GenAI as mentalistic–agentic and anthropomorphic but to express severe critique on the technology's limitations: “*OpenAI's ChatGPT has long been dumb, willing to assist in cybercrime, an Icarus analogy for the age and a threat to sensitive company data*” (ibid). This case is a clear counterexample to framing practices applying anthropomorphising to highlight capabilities, benefits, and possible superiority of LLM-powered GenAI to humans.

Interestingly, some articles deploy mentalistic–agentic framings to LLMs but simultaneously outline why they are not human at all. In these cases, anthropomorphic metaphors serve for sometimes humorous, and sometimes more serious critique:

You can trick AIs like Google Bard, ChatGPT, and Bing AI (which is built on chatGPT) into lying to you, mischaracterizing the truth, and acting like someone who just finished off a bag of mushrooms [...] We expect LLM and AI, which are born of code, algorithms, and never-ending training to be not just smarter than us, but 100% accurate. The flaw, of course, might be that these systems are only as accurate as the training data we've fed them (Appendix 10E).
[...]the most worrying fact to be reiterated is that chatgpt has no commitment to the truth. as the MIT technology review puts it, large language model chatbots are “notorious bullshitters”. (Appendix 10F)

As such, the use of mentalistic–agentic framings that assign anthropomorphic features to LLMs and associated GenAI systems is nuanced, contextual, and serves different rhetorical and argumentative purposes. Relatedly, several articles try to describe them in more technically accurate ways that avoid anthropomorphic metaphors. For example, Singaporean *Strait Times* write in *Generative AI could ease access to justice: High Court judge*: “*generative AI refers to deep-learning algorithms, such as ChatGPT, that can generate high-quality text, images and other content based on the data they were trained on*” (10G). Others explicitly criticise anthropomorphic portrayal: “[...]and if you don't know what ChatGPT is, it can seem overwhelming. So much so that people are starting to project onto ChatGPT, and generative ai in general, qualities and human characteristics that it actually doesn't have” (Appendix 10H).

More technical accounts can still include questionable assumptions about similarities between algorithmic systems and human biological–cognitive characteristics: “*Large language AI models such as LaMDA and the one behind ChatGPT are types of neural networks, which mimic the*

underlying architecture of the brain in computer form. They are fed vast amounts of text from the internet in a process that teaches them how to generate responses to text-based prompts” (Appendix 10I). The problem in this case is that there is no consensus on how the human brain works and neural networks are only one (simplified) attempt at modelling its possible structures. Importantly, the various framing approaches occur often in different articles within the same news outlet, further suggesting that individual journalistic practices are ultimately more decisive than editorial guidelines for AI news coverage, which may not yet exist in many newsrooms.

A noticeable difference between outlets and world regions emerged in the use of the metaphorical framing “human-like”, which can be considered as another conceptual metaphor to explain capabilities and attributes of GenAI that are perceived as similar to human cognition, agency, or behaviour. While in total only 344 articles included the phrase (1.9% of $N_1=17,716$), the sampled news outlets varied visibly in their frequency of including it ($\chi^2[14, N = 17,716] = 112.30, p < 0.001$). Notably, North American outlets such as *The New York Times* and *The Associated Press* included the phrasing more than twice than many other outlets.

That human cognitive capabilities and human roles are generally associated with LLM-driven GenAI systems is further illustrated when exploring the semantic neighbourhoods of specific AI systems. Figs. 6 and 7 show these for ChatGPT and Bard/Gemini. They display frequently co-occurring words with each product name and how these words cluster with each other. For example, in the case of ChatGPT there is a clear cluster around education-related terminology, further illustrating its impact in this domain. Notably, ChatGPT often co-occurs with terms such as “conversation”, “speech”, “skill”, “reasoning” or “human like response”. This looks generally rather similar for Bard/Gemini but without the distinct focus on education-related terminology (Fig. 7).

Notably, for Bard/Gemini, the word clusters imply more focus on user interaction issues, exemplified by terms such as “allow_user” and “feedback” as well as limitations, for example “hallucination”, “mistake”, or “limitation”. This reflects the mixed reception of Bard/Gemini in early 2023, when it was often unfavourably compared to its competitor.

Overall, given the virtually identical value proposition of both services and conceptually same underlying technological infrastructure, these similarities are somewhat expectable. However, they imply that the very purpose for which these technologies were designed—namely, to enable natural conversations as the primary interface for human–computer interaction—strongly predisposes specific framing practices that are prone to anthropomorphic metaphors for conceptual mapping in understanding LLM-driven GenAI.

Fig. 6 Semantic neighbourhood as word clusters for the keyword “ChatGPT” using word embeddings

5 Discussion

The findings of the empirical investigation suggest that news media’s portrayal of LLMs during the sampled period was primarily shaped by the public release of conversational GenAI systems aimed at massive user bases—most notably OpenAI’s ChatGPT. This coverage unfolded alongside reporting on competing commercial offerings, such as Google’s Bard/Gemini. Within the broader field of AI reporting, LLM-centric news forms a distinct sub-strain, in which the novelty of conversational interaction and apparent linguistic competence took centre stage.

Addressing RQ1, which asked how news media portray LLMs and to what extent they evoke agentic, especially mentalistic, framings, the findings point to a complex and nuanced picture. While heavily anthropomorphic framings that reproduce tech-business narratives are highly visible in global AI discourses, they are not necessarily *the* dominant type of LLM and GenAI portrayals. Instead, different framing practices coexist. Agentic mentalistic framing can

serve a variety of communicative goals, including explaining how the technology works conceptually; conveying (and underscoring) the goals and ambitions of technology companies for LLM applications; and providing accessible and engaging-entertaining narratives. Notably, these practices do not per se go hand in hand with a positive-enthusiastic stance towards LLM-driven AI systems but agentic-mentalistic portrayals can serve as means for articulating rather fundamental technology criticism. The use of anthropomorphic framing is highly contextual and cannot be understood as a uniform practice. Crucially, its implications depend on who employs such language and for what purposes. The analysis revealed at least three distinguishable, though not mutually exclusive, modes of usage: 1) uncritical and unreflective adoptions of anthropomorphic metaphors that reproduce technology- and business-driven narratives; 2) more ambiguous or routine uses that do not necessarily amplify AI hype but instead reflect conventional, and often simplified, ways of explaining complex technologies; and 3) adversarial uses, in which anthropomorphic language is

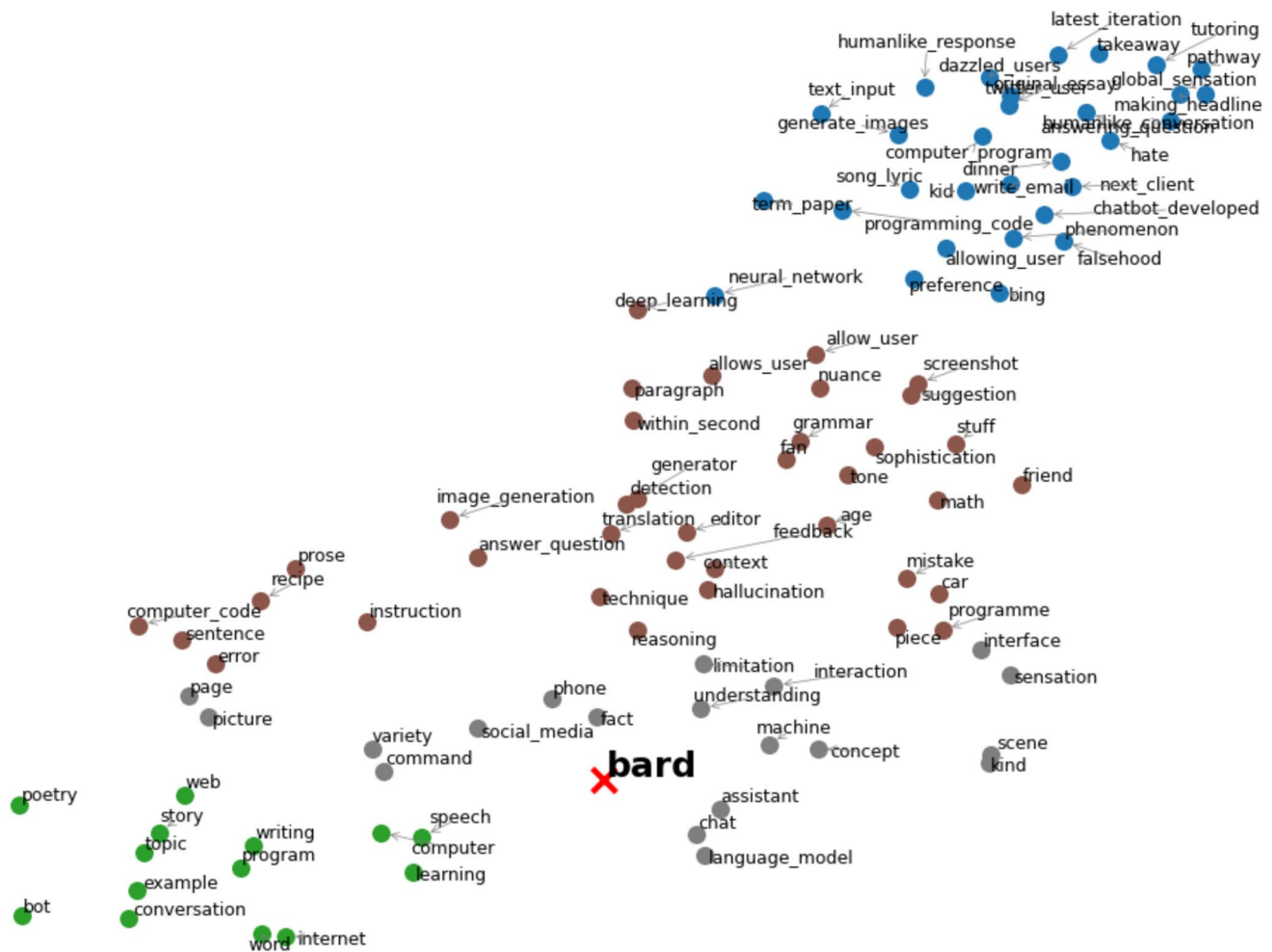


Fig. 7 Semantic neighbourhood as word clusters for the keyword “Bard” (now Gemini) using word embeddings

mobilised to question, critique, or even undermine inflated claims about LLM/GenAI capabilities. These categories are neither exhaustive nor clearly bounded. However, they illustrate that anthropomorphic framing in news discourse cannot be evaluated in a one-dimensional or uniformly negative way. Rather than exclusively reinforcing hype, such framing practices can, in certain contexts, contribute to critically questioning and dispelling technological myths.

However, many articles omit the role of users, prompts, or any form of human agency from their portrayal of GenAI systems, thereby presenting LLMs as autonomous actors rather than as sociotechnical systems embedded in human practices. Articles that take a more “systems-oriented” view mostly discuss business ecosystems in which LLM-based technologies are developed, with limited consideration to political-regulatory factors that shape technology markets. This tendency intersects with broader questions of innovation-centric news framing, where technological capabilities are foregrounded at the expense of contextualisation and critical explanation. At the same time, it stands out

that 45.5% of LLM-related articles present its applications in agentic terms. This suggests that agentic and mentalistic metaphors are perceived as useful journalistic devices.

Regarding RQ2, which examined differences between news outlets based on cultural backgrounds, the results show both convergences as well as difference. While all outlets share similar attention peaks on LLMs, the results reveal uneven contributions to articles framing LLMs as mentalistic–agentic. A small number of outlets—*The New York Times*, *Straits Times*, and *Agence France Presse*—accounted for a disproportionate share of agentic framings. Moreover, observations from Asian news coverage suggest a stronger emphasis on macro-economic and technical developments, rather than on LLM systems themselves. This may partially reflect differences in availability and accessibility of LLM products during the sampled period, particularly in contexts such as China, where coverage often took the form of “look what they have in the West—what does this mean for the economy?”

Turning to RQ3, which focused on differences in framing across specific LLM-based services and products, the

analysis suggests that journalistic discourse, despite nuance, reproduces corporate narratives. In some cases, articles directly cite companies' own mentalistic–agentic framings when describing expanded functionalities, such as the ability to process audio-visual data. This echoes broader critiques of “technochauvinism,” in which technological solutions are presented as self-evidently beneficial and adopted with limited reflection. Particularly salient here is the uncritical or unquestioned uptake of such narratives, which further reinforces agentic portrayals of AI systems. Beyond descriptive patterns, these findings speak to the broader role of news media in shaping public epistemology about AI and the formation of informed publics. While journalism is often understood to fulfil this role for other societal developments, this mandate must be critically assessed considering the specific media ecologies and information economies in which news organisations operate. Media ownership structures, including cases where outlets are controlled by major technology companies, underscore the importance of scrutinising which AI stories are told and how.

Taken together, the analysis maps a range of existing practices and investigates how mentalistic–agentic framings are deployed across contexts. In this regard, engaging with, explaining, or critically reflecting on anthropomorphic discourse is understood as a legitimate journalistic practice, and the presence of such language is not inherently problematic. The analysis is more specifically concerned with instances in which anthropomorphic framing may be adopted uncritically, for example by reproducing technology- or business-driven narratives without sufficient contextualisation. At the same time, the study avoids a one-sided critique of anthropomorphic language, recognising that such expressions can also serve to interrogate and challenge dominant AI imaginaries. The findings are thus intended to provide an empirical basis for more nuanced discussions, informing—rather than prescribing—debates on responsible and critical AI journalism and critical AI literacy. Critical AI literacy has been identified as a key factor shaping contemporary AI news framing practices (Deuze and Beckett 2022). The extent to which journalists understand how emerging technologies work and what societal effects they may have can contribute to more accurate and hype-resistant reporting. From a normative standpoint, critical AI literacy is not a fixed set of skills but rather an ongoing process of deliberation, reflection, and transparency. The ability to critically reflect on one's own profession and the role of AI within it is likely to shape how journalists report on AI for their audiences, who are commonly positioned as the main stakeholders of journalism's public mission. Analysing current news framings of LLMs helps identifying practices that reflect limited critical engagement and highlights areas where AI literacy training may be needed—particularly when linking

framing practices to sociotechnical imaginaries and journalism's role in public epistemology.

Finally, the findings need to be situated against the backdrop of a dual entanglement between news media and GenAI. On the one hand, journalistic and editorial practices are increasingly affected by GenAI systems that promise to automate tasks such as research, copywriting, and content distribution. On the other hand, this development raises critical questions regarding definitions of AI, task allocation, and organisational transparency, while potentially deepening news organisations' dependencies on platform companies. Organisational AI policies, media ownership structures, and individual journalists' attitudes towards emerging technologies all form an important backdrop for agenda-setting and framing choices. Taken together, these dynamics underline the importance of further exploring how different sites of public discourse and opinion formation—news media, corporate communication, and audience perceptions—influence one another.

6 Conclusion

This study set out to examine how news media portray LLMs and to what extent agentic—particularly mentalistic—framings shape contemporary AI reporting. By combining automated, computational techniques with manual content-analytical steps, the study offers a systematic yet interpretative account of LLM-centric news coverage as a distinct, though relatively small, strand within broader AI discourses. The mixed-methods design made it possible to capture both large-scale patterns and more fine-grained framing practices, while situating them within ongoing debates about public epistemology, journalistic responsibility, and critical AI literacy.

Several limitations need to be acknowledged. First, the empirical material was restricted to English-language news content, which necessarily limits the cultural and linguistic scope of the findings. The dataset is also imbalanced in terms of news outlets per world region and future studies should aim for a more even representation, which was not possible due to practical challenges related to data access. Furthermore, several other important factors that distinguish the different news outlets could not be fully explored in the present study, such as their respective political orientation, target audiences, and general news agenda (e.g. mainstream vs. business vs. technology-centric). These variables should be investigated in follow-up studies.

Second, the identification of agentic and mentalistic framings relied on a dictionary-based approach building on Schweitzer et al. (2020). While this enables systematic comparison, it also constrains what can be detected and foregrounds particular linguistic expressions over others.

Relatedly, the labelling of topics and the categorisation of AI-related, LLM-related, and mentalistic–agentic content inevitably involve interpretative decisions that remain subjective and open to debate, despite efforts at transparency and methodological rigour. At the same time, systematically distinguishing between cases in which mentalistic–agentic language uses do or do not evoke mental state attributions falls beyond the scope of this study and would require dedicated psycholinguistic or experimental approaches.

These limitations point directly to avenues for future research. Further studies should engage more explicitly with journalists' own views on and experiences with reporting about AI and LLMs, particularly in relation to expertise, organisational constraints, and perceived audience expectations. In addition, follow-up research would benefit from linguistically diverse data and the development of multilingual methods for comparative text analysis. Importantly, a fully qualitative study should critically unpack the observed framing practices in more detail, especially with a focus on sentiment, rhetorics, and argumentative structures, as the present analysis could only touch upon these. Finally, future comparative studies should more systematically analyse differences and similarities between different LLM-powered AI systems to probe how organisational reputation and brand factors further shape perceptions.

Expanding the empirical scope in these ways would allow for a more robust examination of cultural variation in AI imaginaries and help clarify how different media systems contribute to shaping public understandings of emerging AI technologies.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00146-026-03070-1>.

Author contributions All authors contributed to all aspects of the paper.

Funding No funding was received.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Bareis J, Katzenbach C (2022) Talking AI into being: the narratives and imaginaries of national AI strategies and their performative politics. *Sci Technol Hum Values* 47(5):855–881. <https://doi.org/10.1177/01622439211030>
- Bender EM, Hanna A (2025) The AI Con: How to fight big tech's hype and create the future we want. Harper
- Blut M, Wang C, Wunderlich NV, Brock C (2021) Understanding anthropomorphism in service provision: a meta-analysis of physical robots, chatbots, and other AI. *J Acad Mark Sci* 49(4):632–658
- Brandmayr F (2021) Public epistemologies and intellectual interventions in contemporary Italy. *Int J Polit Cult Soc* 34:47–68. <https://doi.org/10.1007/s10767-019-09346-3>
- Bunz M, Braghieri M (2021) The AI doctor will see you now: assessing the framing of AI in news coverage. *AI Soc*. <https://doi.org/10.1007/s00146-021-01145-9>
- Chong D, Druckman JN (2007) Framing theory. *Annu Rev Polit Sci* 10:103–126. <https://doi.org/10.1146/annurev.polisci.10.072805.103054>
- Chuan C-H (2023) A critical review of news framing of artificial intelligence. In: Lindgren S (ed) *Handbook of critical studies of artificial intelligence*. Edward Elgar Publishing, pp 266–276
- Ciriello R, Chen A, Rubinsztajn Z (2025) Think Miss Piggy, not Pinocchio: debunking the myth of 'autonomous' AI. *AI Soc*. <https://doi.org/10.1007/s00146-025-02252-7>
- Cools H, Van Gorp B, Opgenhaffen M (2022) Where exactly between utopia and dystopia? A framing analysis of AI and automation in US newspapers. *Journalism* 25(1):3–21. <https://doi.org/10.1177/14648849221122647>. (Original work published 2024)
- Cordeschi R (2007) AI turns fifty: revisiting its origins. *Appl Artif Intell* 21(4–5):259–279
- Crépel M, Do S, Cointet J-P, Cardon D, Bouachera Y (2021) Mapping AI issues in media through NLP methods. *CHR2021: computational Humanities Research Conference*, November 2021, Amsterdam
- Deuze M, Beckett C (2022) Imagination, algorithms and news: developing AI literacy for journalism. *Digit J* 10(10):1913–1918. <https://doi.org/10.1080/21670811.2022.2119152>
- Emmert-Streib F, Yli-Harja O, Dehmer M (2020) Artificial intelligence: a clarification of misconceptions, myths and desired status. *Front Artif Intell* 3:524339
- Epley N, Waytz A, Cacioppo JT (2007) On seeing human: a three-factor theory of anthropomorphism. *Psychol Rev* 114(4):864
- Epstein Z, Levine S, Rand DG, Rahwan I (2020) Who gets credit for AI-generated art? *iScience*. <https://doi.org/10.1016/j.isci.2020.101515>
- Fischer RA-L, Walczuch R, Guzman E (2021) Does culture matter? Impact of individualism and uncertainty avoidance on app reviews. *arXiv*. <https://doi.org/10.48550/arXiv.2102.06311>. (arXiv:2102.06311v2)
- Gray HM, Gray K, Wegner DM (2007) Dimensions of mind perception. *Science* 315(5812):619–619
- Grootendorst M (2022) BERTopic Neural topic modeling with a class-based TF-IDF procedure. *arXiv Preprint*. <https://doi.org/10.48550/arXiv.2203.05794>
- Gutierrez S, Nguyen D, van Es K (2025) Tool, companion or a catalyst force? Exploring sociotechnical imaginaries within AI livestreams' communities of practice. *Big Data Soc*. <https://doi.org/10.1177/20539517251381>
- Hakken D (1993) Computing and social change: new technology and workplace transformation, 1980–1990. *Annu Rev Anthropol* 22:107–132
- Honnibal M, Montani I, Van Landeghem S, Boyd A (2020) spaCy: Industrial-strength natural language processing in Python [Software]. Zenodo. <https://doi.org/10.5281/zenodo.1212303>
- Ittefaq M, Zain A, Arif R, Ala-Uddin M, Ahmad T, Iqbal A (2025) Global news media coverage of artificial intelligence (AI): a

- comparative analysis of frames, sentiments, and trends across 12 countries. *Telemat Inform* 96:102223. <https://doi.org/10.1016/j.tele.2024.102223>
- Jasanoff S (2015) Future imperfect: science, technology, and the imaginations of modernity. In: Jasanoff S, Kim SH (eds) *Dreamscapes of modernity: sociotechnical imaginaries and the fabrication of power*. The University of Chicago Press, Chicago, pp 1–33
- Johnson DG, Verdicchio M (2017) Reframing AI discourse. *Mind Mach* 27(4):575–590
- Jones B, Jones R, Luger E (2022) AI ‘everywhere and nowhere’: addressing the AI intelligibility problem in public service journalism. *Digit J* 10(10):1731–1755. <https://doi.org/10.1080/21670811.2022.2145328>
- Kelly JP (2009) Not so revolutionary after all: the role of reinforcing frames in US magazine discourse about microcomputers. *New Media Soc* 11(1–2):31–52
- Kotliar DM (2026) Can’t stop the hype: scrutinizing AI’s realities. *Inf Commun Soc*. <https://doi.org/10.1080/1369118X.2025.2531165>
- Kövecses Z (2004) Introduction: cultural variation in metaphor. *Eur J Engl Stud* 8(3):263–274. <https://doi.org/10.1080/1382557042000277386>
- Krennmayr T (2014) What corpus linguistics can tell us about metaphor use in newspaper texts. *J Stud* 16(4):530–546. <https://doi.org/10.1080/1461670X.2014.937155>
- Kvasny L, Truex D (2001) Defining away the digital divide: a content analysis of institutional influences on popular representations of technology. In: Russo NL, Fitzgerald B, DeGross JI (eds) *Realigning Research and Practice in Information Systems Development*. IFIP—The International Federation for Information Processing. Springer, Boston, MA
- Lakoff G, Johnson M (1981) *Metaphors we live by*. University of Chicago Press, Chicago
- Lammar D, Horst M, Müller R (2025) AI in the German media: narratives of AI-in-particular and AI-in-general in German media reporting about artificial intelligence. *Digit J*. <https://doi.org/10.1080/21670811.2025.2493759>
- Langer, M., Hunsicker, T., Feldkamp, T., König, C. J., Grgić-Hlača, N. (2022) “Look! It’s a computer program! It’s an algorithm! It’s AI!”: Does terminology affect human perceptions and evaluations of algorithmic decision-making systems?. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (pp. 1–28).
- Lewis SC, Zamith R, Bunquin JBA (2025) Technological hype and AI in journalism: five functions and why they matter. *Digit J*. <https://doi.org/10.1080/21670811.2025.2557994>
- Li M, Suh A (2022) Anthropomorphism in AI-enabled technology: a literature review. *Electron Mark* 32(4):2245–2275
- Liebherr M, Babiker A, Alshakhsi S et al (2025) Artificial intelligence vs. users’ well-being and the role of personal factors: a study on Arab and British samples. *J Technol Behav Sci* 10:811–820. <https://doi.org/10.1007/s41347-025-00503-4>
- Liu F, Makady H, Nah S, McNealy J (2023) When citizens support AI policies: the moderating roles of AI efficacy on AI news, discussion, and literacy. *J Inf Technol Polit* 21(4):493–509. <https://doi.org/10.1080/19331681.2023.2294363>
- Mikolov, T., Chen, K., Corrado, G., Dean, J. (2013) Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*. <https://arxiv.org/abs/1301.3781>
- Minsky M (2007) *The emotion machine: Commonsense thinking, artificial intelligence, and the future of the human mind*. Simon and Schuster
- Nass, C., Steuer, J., Tauber, E., Reeder, H. (1993) Anthropomorphism, agency, and ethopoeia: computers as social actors. In *INTERACT’93 and CHI’93 conference companion on Human factors in computing systems* (pp. 111–112).
- Nguyen D (2023) How news media frame data risks in their coverage of big data and AI. *Intern Policy Rev*. <https://doi.org/10.14763/2023.2.1708>
- Nguyen D, Hekman E (2022) A ‘new arms race’? Framing China and the U.S.A. in A.I. news reporting: a comparative analysis of the Washington Post and South China Morning Post. *Glob Media China* 7(1):58–77. <https://doi.org/10.1177/20594364221078626>
- Nguyen D, Hekman E (2024) The news framing of artificial intelligence: a critical exploration of how media discourses make sense of automation. *AI & Soc* 39(2):437–451. <https://doi.org/10.1007/s00146-022-01511-1>
- Ouchchy L, Coin A, Dubljević V (2020) AI in the headlines: the portrayal of the ethical issues of artificial intelligence in the media. *AI Soc* 35(4):927–936
- Quesque F, Apperly I, Baillargeon R, Baron-Cohen S, Becchio C, Bekkering H, Brass M (2024) Defining key concepts for mental state attribution. *Commun Psychol* 2(1):29
- Rahman MM, Babiker A, Ali R (2025) Motivation, concerns, and attitudes towards ai differences by gender, age, and culture. In: Barhamgi M, Wang H, Wang X (eds) *Web Information Systems Engineering—WISE 2024*. WISE 2024. Lecture Notes in Computer Science. Springer, Singapore
- Roe J, Perkins M (2023) ‘What they’re not telling you about Chat-GPT’: exploring the discourse of AI in UK news media headlines. *Humanit Soc Sci Commun* 10:753. <https://doi.org/10.1057/s41599-023-02282-w>
- Roesler E, Manzey D, Onnasch L (2021) A meta-analysis on the effectiveness of anthropomorphism in human-robot interaction. *Sci Robot* 6(58):eabj5425
- Saldaña, J. (2021) *The coding manual for qualitative researchers* (4th ed.). SAGE.
- Sartori L, Bocca G (2023) Minding the gap(s): public perceptions of AI and socio-technical imaginaries. *AI & Soc* 38:443–458. <https://doi.org/10.1007/s00146-022-01422-1>
- Schweitzer S, Waytz A (2021) Language as a window into mind perception: how mental state language differentiates body and mind, human and nonhuman, and the self from others. *J Exp Psychol Gen* 150(8):1642–1672. <https://doi.org/10.1037/xge0001013>
- Shanahan M (2024) Talking about large language models. *Commun ACM* 67(2):68–79
- Shorey S, Rodriguez E (2025) Automating essential work: news media and technology hype in a critical sector 2014–2024. *Digit J*. <https://doi.org/10.1080/21670811.2025.2513411>
- Stahl WA (1995) Venerating the black box: Magic in media discourse on technology. *Sci Technol Hum Values* 20(2):234–258
- Sun S, Zhai Y, Shen B, Chen Y (2020) Newspaper coverage of artificial intelligence: a perspective of emerging technologies. *Telemat Inform* 53:101433. <https://doi.org/10.1016/j.tele.2020.101433>
- Valderrama Barragán M, Tironi M, Cotoras D, Correa T, Humeres M, López C (2025) From industry hype to emerging criticism: analysing Chilean news media coverage of artificial intelligence. *Digit J*. <https://doi.org/10.1080/21670811.2025.2450624>
- Waytz A, Cacioppo J, Epley N (2010) Who sees human? The stability and importance of individual differences in anthropomorphism. *Perspect Psychol Sci* 5(3):219–232. <https://doi.org/10.1177/1745691610369336>
- Wischniewski M, Ngo T, Bernemann R, Jansen M, Krämer N (2022) “I agree with you, bot!” how users (dis)engage with social bots on Twitter. *New Media Soc* 26(3):1505–1526. <https://doi.org/10.1177/14614448211072307>. (Original work published 2024)
- Zeng J, Chan C, Schäfer MS (2022) Contested Chinese dreams of AI? Public discourse about artificial intelligence on WeChat and People’s Daily online. *Inf Commun Soc* 25(3):319–340